# THE HILBERT-POLYNOMIAL PIECE OF QUOT IS SEPARATED OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 1269-1281 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 1283-1339, the complete author proof (55 body lines).

*Editorial note.* Navigation stubs: 13 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 4 of queue_moduli.json (pb=2455, stmt 1269-1281, proof 1283-1339).

*Editorial note.* stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma 0DP3: sources/stacks-moduli/pinned-original-source.tex, source lines 870–881. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0DP8: sources/stacks-moduli/pinned-original-source.tex, source lines 1018–1027. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0DPC: sources/stacks-moduli/pinned-original-source.tex, source lines 1233–1244. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0DM3: sources/stacks-moduli/pinned-original-source.tex, source lines 755–759. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 01RG: sources/stacks-moduli/pinned-original-source.tex, source lines 1339–1339. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0G41: sources/stacks-moduli/pinned-original-source.tex, source lines 1339–1339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0DP6: sources/stacks-moduli/pinned-original-source.tex, source lines 950–981. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0D31: sources/stacks-moduli/pinned-original-source.tex, source lines 1339–1339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 07SK: sources/stacks-moduli/pinned-original-source.tex, source lines 1339–1339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0D2X: sources/stacks-moduli/pinned-original-source.tex, source lines 1339–1339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 07V7: sources/stacks-moduli/pinned-original-source.tex, source lines 1339–1339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 084T: sources/stacks-moduli/pinned-original-source.tex, source lines 1339–1339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 03KG: sources/stacks-moduli/pinned-original-source.tex, source lines 1339–1339. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $f : X \to B$ be a separated morphism of finite presentation of algebraic spaces. Let $\mathcal{F}$ be a finitely presented $\mathcal{O}_X$-module. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module ample on $X/B$, see Divisors on Spaces, Definition 0D31. The algebraic space $\mathrm{Quot}^P_{\mathcal{F}/X/B}$ parametrizing quotients of $\mathcal{F}$ having Hilbert polynomial $P$ with respect to $\mathcal{L}$ is separated of finite presentation over $B$.*

*Proof.* We have already seen that $\mathrm{Quot}_{\mathcal{F}/X/B} \to B$ is separated and locally of finite presentation, see Lemma 0DM3. Thus it suffices to show that the open subspace $\mathrm{Quot}^P_{\mathcal{F}/X/B}$ of Remark 0DP6 is quasi-compact over $B$.

The question is étale local on $B$ (Morphisms of Spaces, Lemma 03KG). Thus we may assume $B$ is affine.

Assume $B = \mathrm{Spec}(\Lambda)$. Write $\Lambda = \mathrm{colim}\, \Lambda_i$ as the colimit of its finite type $\mathbf{Z}$-subalgebras. Then we can find an $i$ and a system $X_i, \mathcal{F}_i, \mathcal{L}_i$ as in the lemma over $B_i = \mathrm{Spec}(\Lambda_i)$ whose base change to $B$ gives $X, \mathcal{F}, \mathcal{L}$. This follows from Limits of

Spaces, Lemmas 07SK (to find $X_i$), 07V7 (to find $\mathcal{F}_i$), 0D2X (to find $\mathcal{L}_i$), and 084T (to make $X_i$ separated). Because

$$\mathrm{Quot}_{\mathcal{F}/X/B} = B \times_{B_i} \mathrm{Quot}_{\mathcal{F}_i/X_i/B_i}$$

and similarly for $\mathrm{Quot}^P_{\mathcal{F}/X/B}$ we reduce to the case discussed in the next paragraph.

Assume $B$ is affine and Noetherian. We may replace $\mathcal{L}$ by a positive power, see Lemma 0DP8. Thus we may assume there exists an immersion $i : X \to \mathbf{P}^n_B$ such that $i^*\mathcal{O}_{\mathbf{P}^n}(1) = \mathcal{L}$. By Morphisms, Lemma 01RG there exists a closed subscheme $X' \subset \mathbf{P}^n_B$ such that $i$ factors through an open immersion $j : X \to X'$. By Properties, Lemma 0G41 there exists a finitely presented $\mathcal{O}_{X'}$-module $\mathcal{G}$ such that $j^*\mathcal{G} = \mathcal{F}$. Thus we obtain an open immersion

$$\mathrm{Quot}_{\mathcal{F}/X/B} \longrightarrow \mathrm{Quot}_{\mathcal{G}/X'/B}$$

by Lemma 0DP3. Clearly this open immersion sends $\mathrm{Quot}^P_{\mathcal{F}/X/B}$ into $\mathrm{Quot}^P_{\mathcal{G}/X'/B}$. Now $\mathrm{Quot}^P_{\mathcal{G}/X'/B}$ is proper over $B$ by Lemma 0DPC. Therefore it is Noetherian and since any open of a Noetherian algebraic space is quasi-compact we win. □